



Open-Motion

User Manual — Clinical build

Product	Open-Motion blood-flow monitor
Application version	1.5.3 (Clinical build)
Compatible firmware	Console 1.8.1 · Sensor modules 1.8.2
Document revision	September 2026 — draft
Audience	Operators

DRAFT — not a controlled document. This manual is a documentation preview written against Open-Motion 1.5.3 and illustrated with screenshots of the 1.5.3 Clinical build running on bench hardware. It is not a validated Instructions-For-Use document — always defer to official Openwater labeling and training.

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1. About the system

Open-Motion is Openwater's optical blood-flow monitor. Two forehead **sensor modules** (eight cameras each) and a **console** containing a near-infrared laser measure laser-speckle images through the skin and compute, in real time:

- **BFI — Blood Flow Index:** a relative index of blood flow (white trace).
- **BVI — Blood Volume Index:** a relative index of blood volume (blue trace). BVI is smoothed with a low-pass filter, which is always on in the Clinical build.

The **Open-Motion** application presents a simple, guided workflow: connect, check sensor contact, scan, review. Each scan uses four cameras per sensor module (cameras 1, 2, 7 and 8, shown as the **Far** configuration). The display shows large LEFT and RIGHT BFI and BVI readouts — the average of that side's active cameras — beside one live plot per side. Every scan starts with an automatic check that the sensors are seated against the skin.

Laser safety is enforced by a hardware interlock inside the console, independently of this software.

2. Installing and starting

2.1 Installation and updates

- Open-Motion is installed with the installer supplied by Openwater. The installer and the application are code-signed by Openwater, so Windows names Openwater as the publisher during installation.
- The Clinical build has **no in-app updater**. To move to a newer version, install the new release supplied by Openwater.
- **Each Windows account has its own scan history, settings and logs** (see §10). Sign in to Windows with the same account each time so earlier scans remain visible.
- **Upgrading from 1.5.2 or earlier:** the installed application starts with an empty scan history and default display preferences. Earlier scans are not deleted — they remain in `C:\ProgramData\Openwater\data` — but this version does not show them.

2.2 Starting the application

Launch **Open-Motion** from the Start menu or the desktop shortcut. The application is a single signed file that unpacks itself on launch, so the window takes a few seconds to appear.

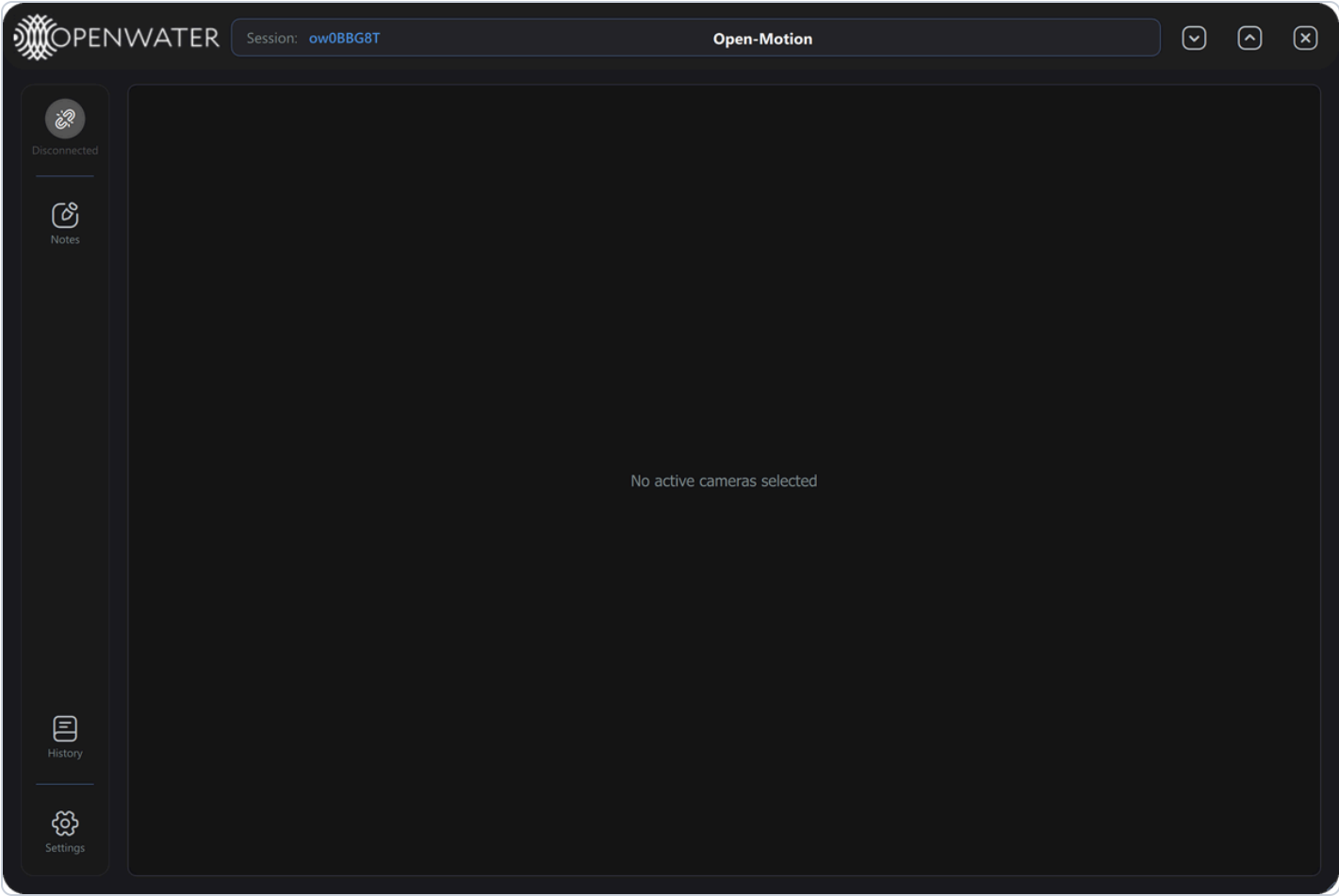
- Only one copy can run at a time. A second launch shows *"Another instance of the application is already running. Please close the existing instance before opening a new one."* — close the other copy first.
- The application connects to the console and the sensor modules **automatically** whenever they are plugged in and powered on. There is no Connect button.
- About 12 seconds after launch the application checks what it found. If a device is missing, a yellow message appears in the bottom-right corner:
 - *"Console not detected. Check the console USB cable and power, then reconnect."*
 - *"Sensor not detected. Check the sensor USB cable and power, then reconnect."*
 - *"System not found. Check that the console and sensor are connected and powered on."* (nothing was found)
- After a sensor module connects, the system initializes it for a few seconds. The **Start** label stays dimmed and cannot be pressed until initialization finishes.

2.3 Connection states

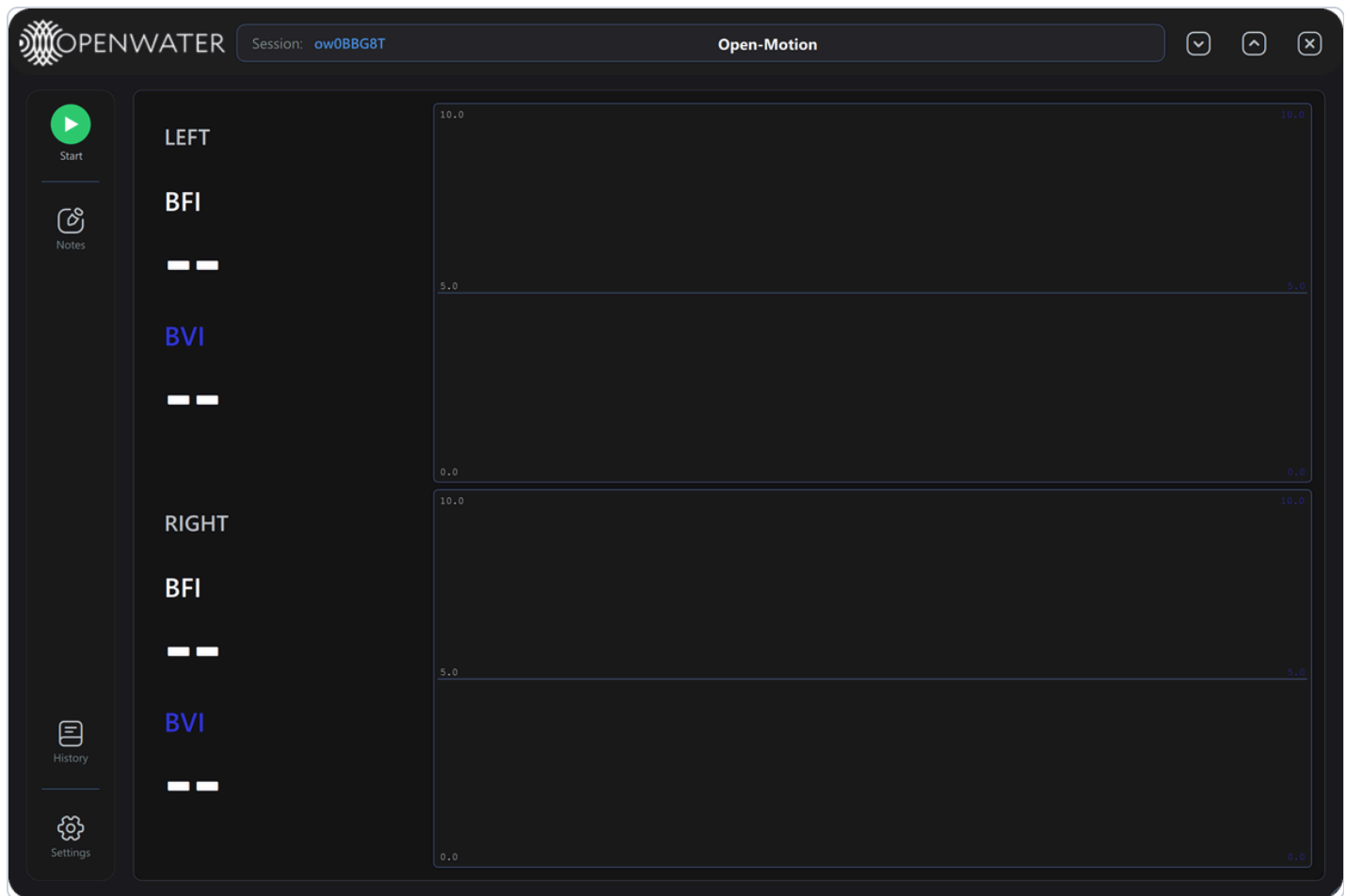
The round badge at the top of the left toolbar shows the system state:

Badge	Label	Meaning
Grey circle, chain-link icon	Disconnected	The console or both sensor modules are not connected. Scanning is disabled.

Badge	Label	Meaning
Green circle, dimmed label	Start	Connected; a sensor module is still initializing. Wait a few seconds.
Green circle, play icon	Start	Ready to scan.
Yellow circle	Start	Start was pressed; the system is finishing the previous step before the scan begins (normally a few seconds).
Red circle, stop icon	Stop	A scan is running. Press to stop it.



Disconnected: the grey badge reads "Disconnected" and the plot area shows "No active cameras selected".

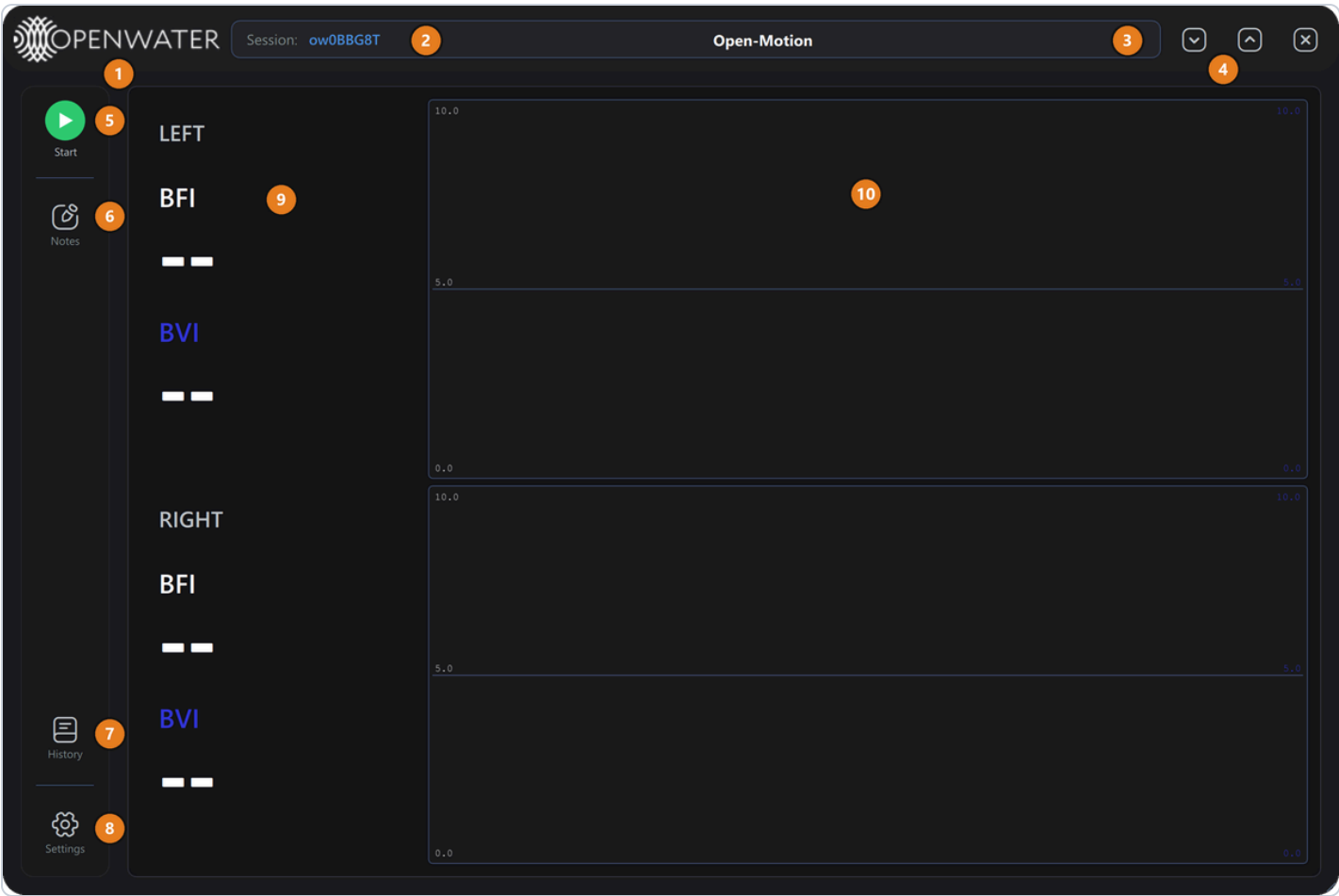


Ready: green Start badge; the LEFT and RIGHT panels show -- until a scan produces data.

2.4 Console indicator light

Console light	Meaning
Solid green	Powered and idle.
Solid blue	The laser is active (during a contact-quality check or a scan).
Blinking blue	An error. The application shows an error dialog (§9.3); the light returns to green once the dialog is dismissed. If the light blinks with no dialog in the application, power-cycle the console.

3. The main screen



#	Element	What it does
1	Openwater logo	Branding.
2	Session	The session label, e.g. <code>ow0BBG8T</code> . A new label is generated each time the application starts and is stored with every scan recorded during that session.
3	Scan clock	Blank while idle. During a scan it shows the elapsed time <code>HH:MM:SS</code> in green.
4	Window controls	Minimize, maximize/restore, and close.
5	Start / Stop badge	Starts the scan workflow (contact-quality check, then scan) or stops the running scan. Disabled until the system is ready (§2.3).
6	Notes	Opens Session Notes (§6). Always available.
7	History	Opens Scan History — review, replay, export or delete past scans (§7). Disabled during a scan.
8	Settings	Opens Settings (§8). Disabled during a scan.

#	Element	What it does
9	LEFT / RIGHT panels	Large live BFI and BVI readouts for each side (§5.1).
10	Plots	One plot per side with the BFI (white) and BVI (blue) traces. The timeline bar and window-length control appear under the plots once a scan has been recorded (§5).

Window behavior:

- **Move** the window by dragging the header bar.
- **Resize** with the diagonal grip in the bottom-right corner (minimum 800 × 600).
- **Close while busy:** if you press  during a scan or contact-quality check, the application does not exit immediately. A yellow message such as "Scan in progress. Click X again to cancel and exit." appears. Press  again within 5 seconds to cancel the work and exit.

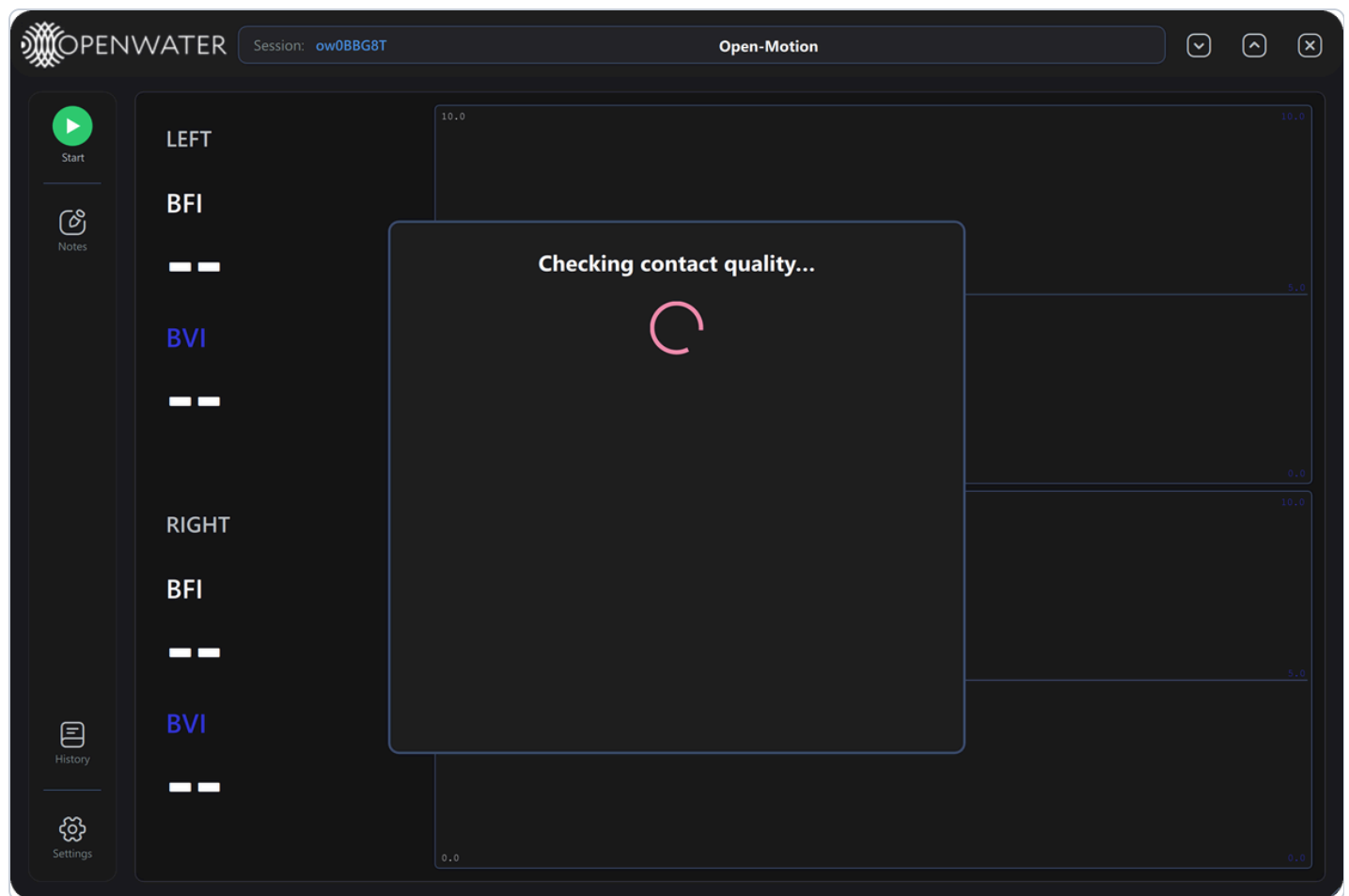
4. Running a scan

Step 1 — Place the sensors

Place the sensor modules symmetrically about the midline on the patient's forehead, in direct contact with the skin, above the brow line, with no obstructions or debris between the sensor modules and the skin. (The same instructions are in Settings → Sensor Placement Instructions.)

Step 2 — Press Start

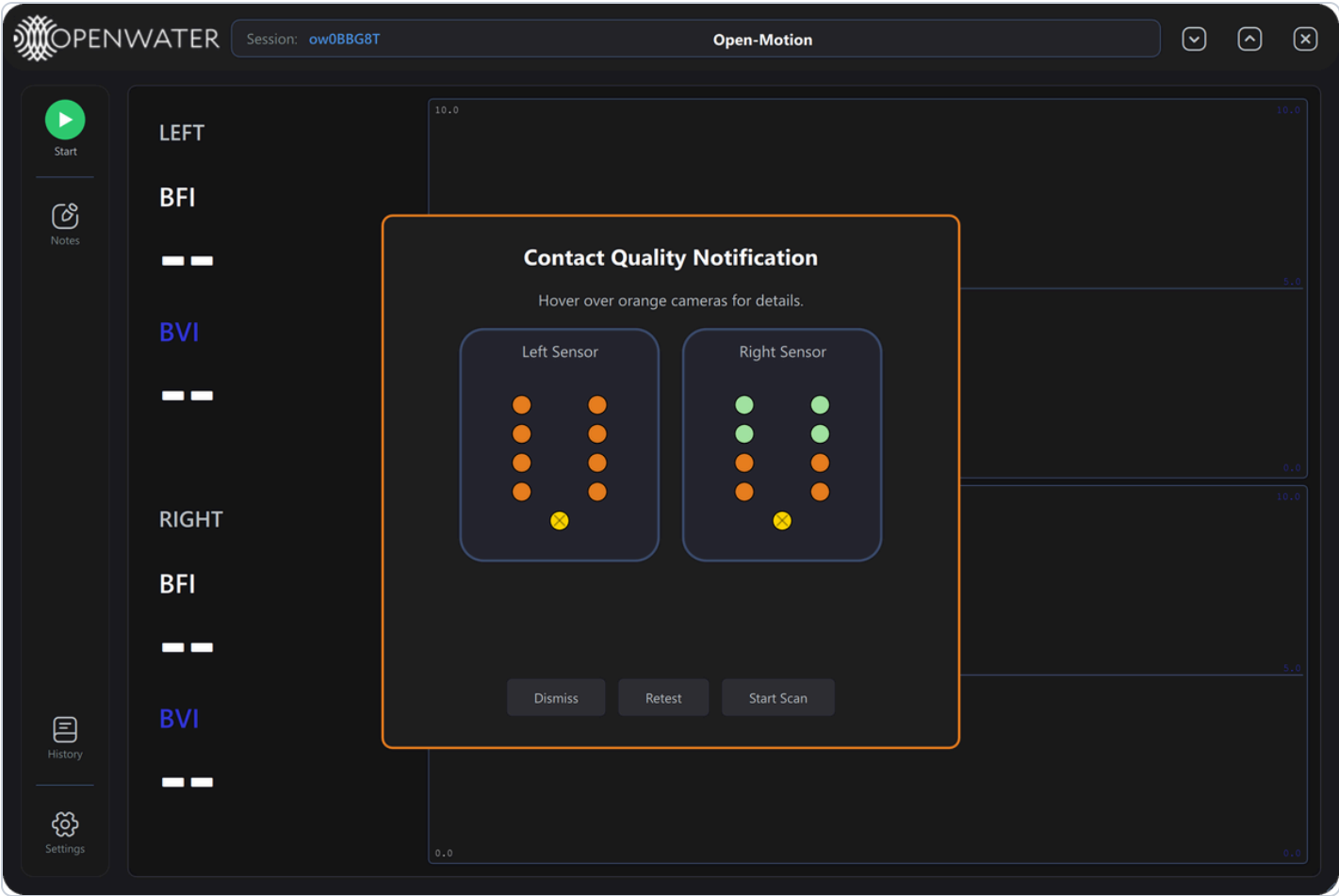
Press the green **Start** badge. Every scan begins with an automatic **contact-quality check**. The laser is on during the check.



"Checking contact quality..." — the check takes about 10 seconds.

The check tests **all eight cameras** on each connected sensor module — including the cameras a scan does not use — so it confirms that the whole module is seated. The check dialog cannot be closed by clicking outside it; use its buttons.

Step 3 — Review the contact-quality result



A result with warnings. Here the left module was not in contact, and the right module had good contact on its top four cameras only.

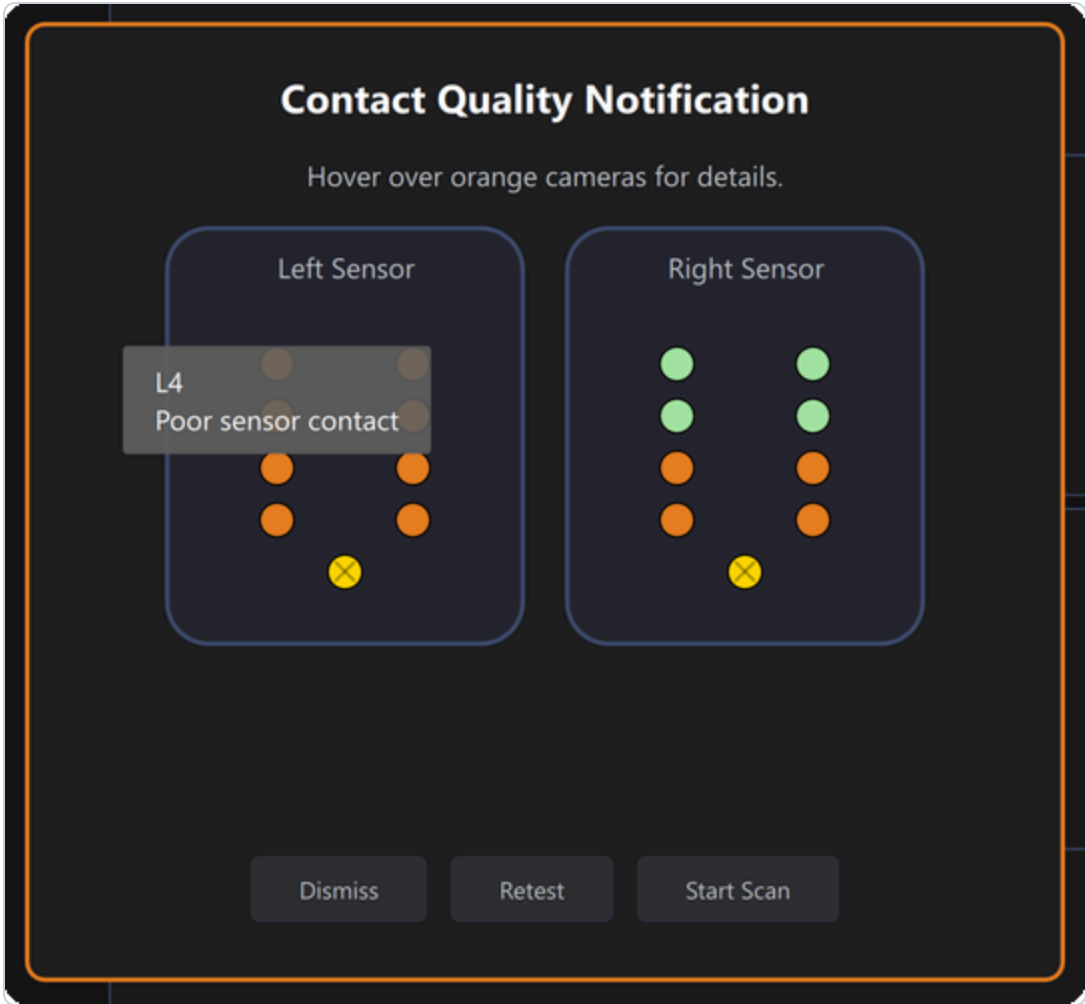
The border color and title summarize the result:

Border	Title	Meaning
Green	Good signal quality	All cameras report acceptable ambient light and contact levels.
Orange	Contact Quality Notification	One or more cameras have a problem. Hover over an orange camera for the reason.
Red	Contact check failed	The check could not be completed; the reason is shown in red (for example, a device disconnected).

Each sensor diagram shows the eight cameras in the module's U shape — cameras 1 to 4 down the left column and 8 to 5 down the right column — with the laser aperture (yellow ⊗) at the bottom:

Camera dot	Meaning
Green	Good contact.

Camera dot	Meaning
Orange	Problem detected — hover for the reason.
Grey	Not evaluated (still checking, or not part of the running scan).



Hovering over a camera shows its label (L = left module, R = right module, then the camera number) and the problem found.

Reason	What to do
Poor sensor contact	The camera is not receiving enough light back from the skin. Re-seat the module flat against the skin and check for hair or debris.
Ambient light detected	Room light is reaching the camera. Improve the seal against the skin or shade the sensors from bright light.

Buttons:

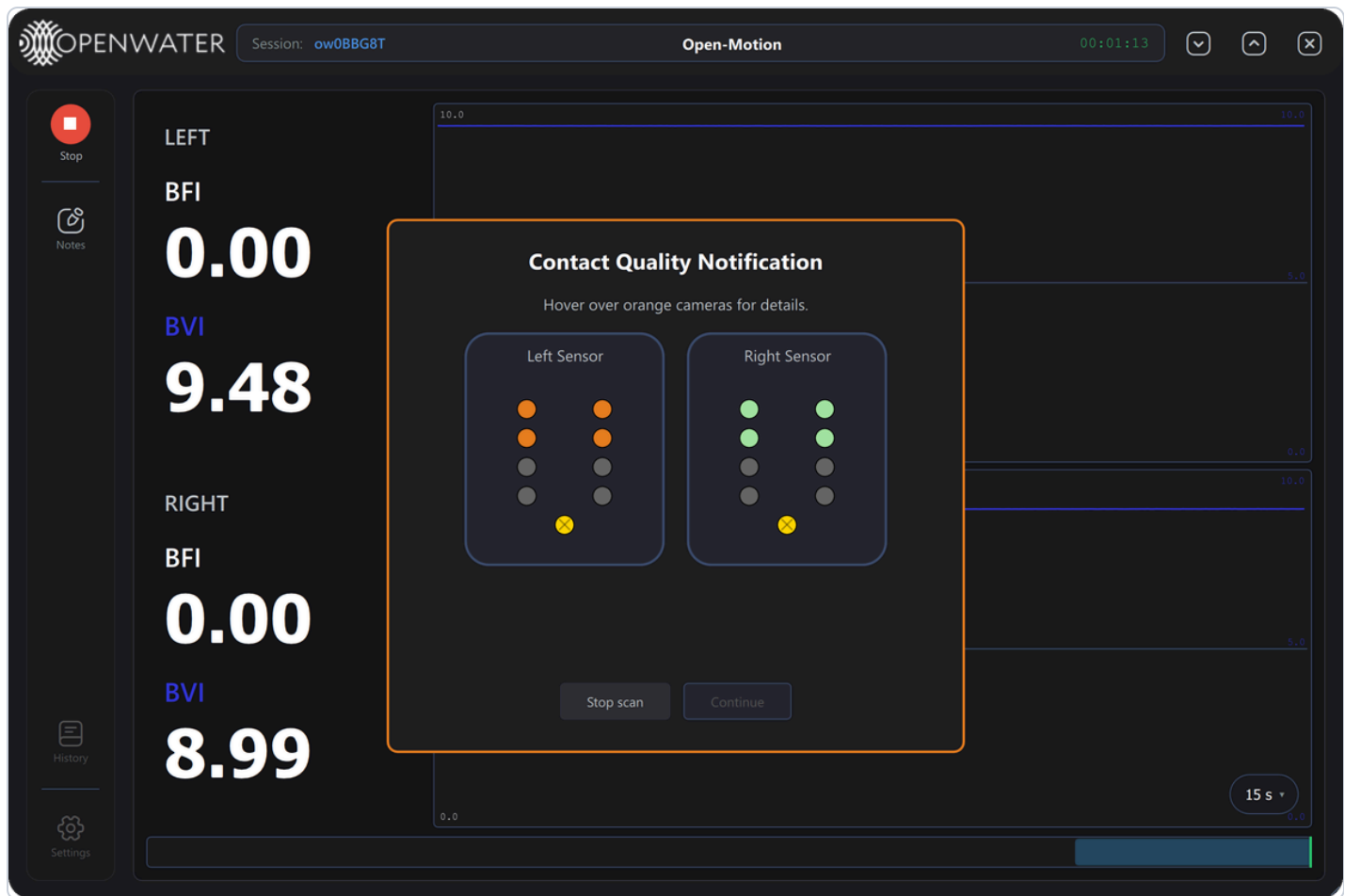
Button	Action
Dismiss	Close the dialog without scanning.
Retest	Run the check again (after adjusting the sensors).
Start Scan	Start the scan. Available as soon as the check has finished, whatever the result — for best data quality, resolve orange cameras before starting.

Step 4 — During the scan

After **Start Scan** the badge may turn yellow for a moment, then red (**Stop**), and the header clock starts counting. The LEFT and RIGHT panels show live BFI and BVI values and the plots draw the traces.

- The scan is **open-ended**: it runs until you press **Stop**, up to a maximum of 12 hours.
- **History** and **Settings** are disabled during a scan. **Notes** stays available.
- Press **Space** to open Session Notes with a timestamp already inserted (§6).
- The console light is solid blue while the laser is on.

Contact warnings during a scan. If contact degrades while scanning, the contact-quality dialog reappears on top of the plots:



A contact warning during a scan. Only the four scan cameras per side are evaluated; the others are grey. The scan keeps recording while the dialog is shown.

Button	Action
Stop scan	End the scan now.
Continue	Return to the plots. Enabled only after every problem has stayed clear for a few seconds — re-seat the sensor and wait for the border to turn green and the message "All contact quality issues are currently inactive. You may dismiss."

While problems remain active, the dialog stays open until they clear or the scan is stopped (with **Stop scan** or the red **Stop** badge in the toolbar).

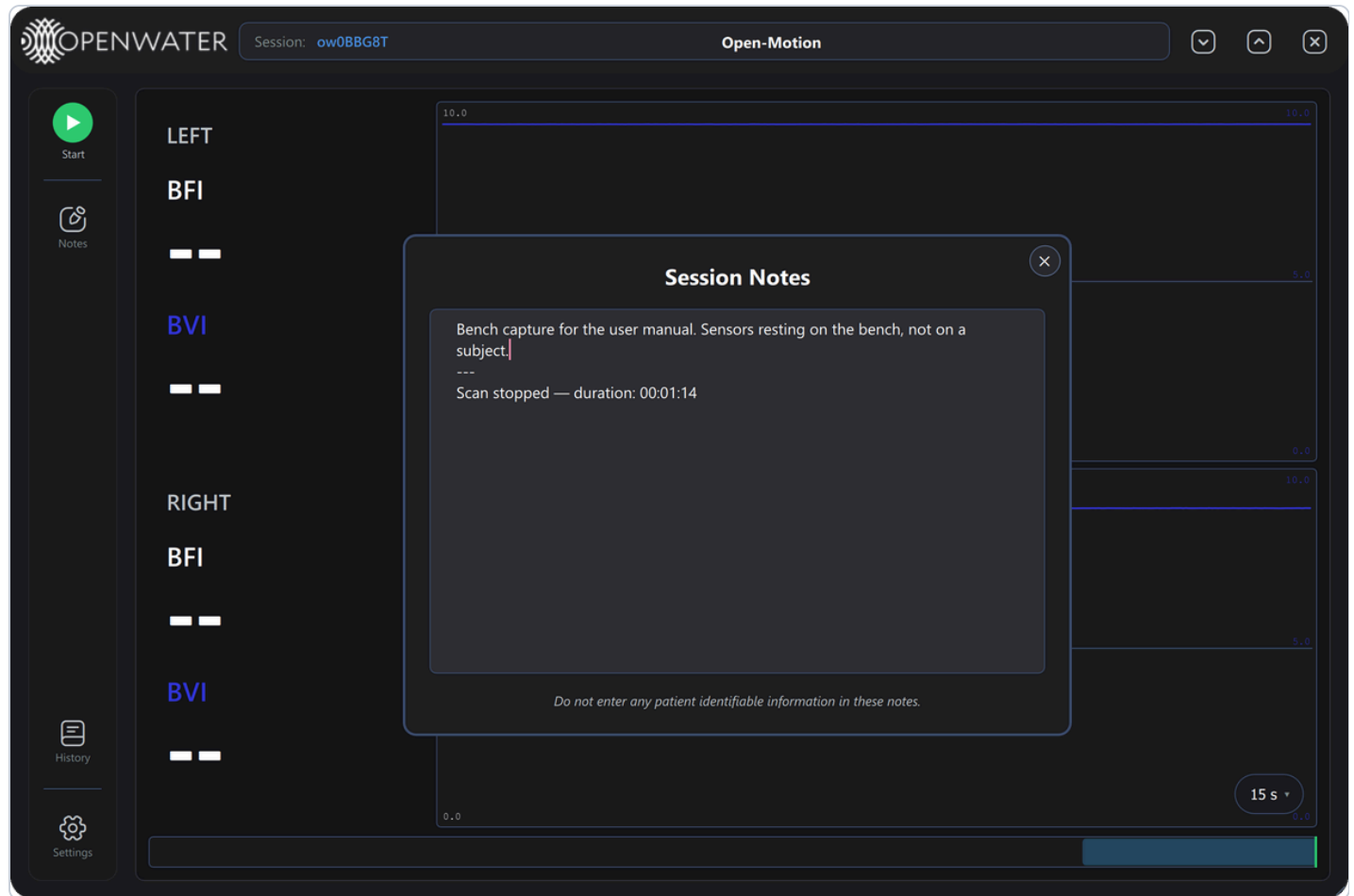
Camera messages. Short messages in the bottom-right corner report individual cameras:

- "Camera LEFT 2 connection lost at 00:01:23 — last temp 41.2°C" — that camera has delivered no data for more than 2 seconds. The scan continues on the remaining cameras. Check the sensor module's cable and power; if cameras keep dropping out, stop the scan and contact support.
- "Camera LEFT 2 reconnected at 00:01:40" — data from that camera resumed.

If every camera stops delivering data, the scan is stopped automatically (error E-303, \$9.4).

Step 5 — Stop and annotate

Press the red **Stop** badge (or **Stop scan** in a contact warning). After a few seconds of shutdown, **Session Notes** opens automatically with a record of the scan:



The stop line and the duration are filled in automatically ("Scan stopped — duration: 00:01:14"). Type any observations, then close the window with X to save them.

- If the application detected gaps longer than 1 second in the data, a "Data gaps (> 1.0s): ..." line is added under the duration.
- All scan data is saved automatically to the application's encrypted database. Nothing needs to be exported for the data to be retained.
- If a scan ends unexpectedly, a message reports what was kept:
 - "Scan ended unexpectedly — partial data was saved. The final segment could not be dark-corrected and was discarded." (yellow)
 - "Scan ended unexpectedly and no data was recorded (the device may have disconnected mid-scan). This scan was not saved." (red; stays on screen until dismissed)

5. Reading and navigating the plots

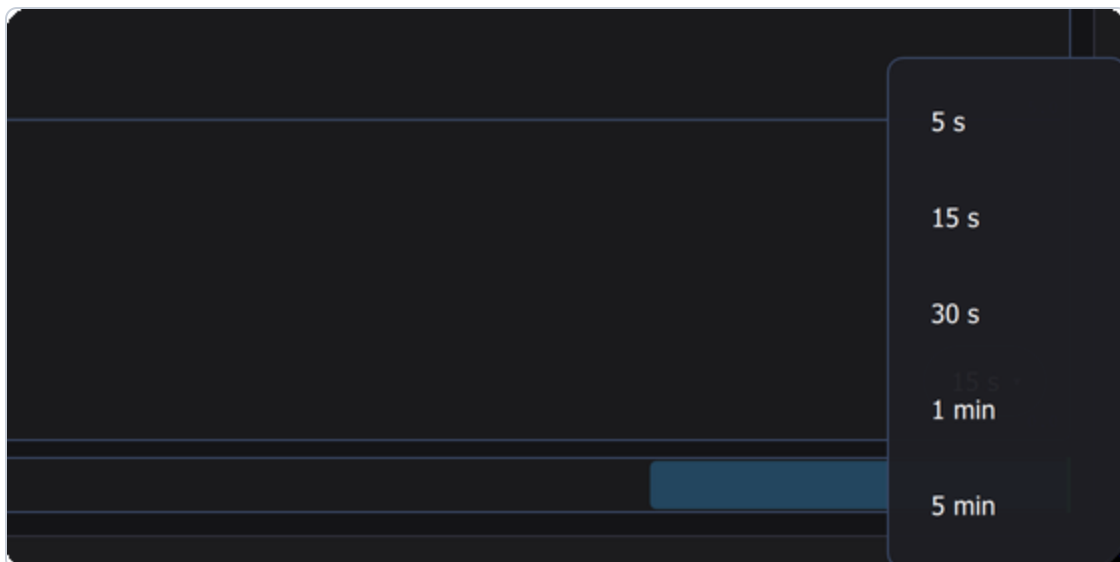
5.1 The readouts

- Each side has one plot with two traces: **BFI** (white) and **BVI** (blue). The vertical axis is fixed to the Manual Plot Bounds in Settings (default 0 to 10 for both).
- The large numbers are the side average of the active cameras at the latest time shown. Readings are limited to the 0–10 display range: a value below 0 shows as `0.00` and a value above 10 shows as `10.00`.
- `--` means there is no valid reading — no scan has run yet, the scan has stopped, or the data at that moment was invalid.

5.2 Navigation

The plot area works like a video recorder: you can rewind during or after a scan without losing the live recording.

Control	Where	What it does
Window-length pill (e.g. <code>15 s ▾</code>)	Bottom-right of the plots	Chooses how much time is visible: 5 s, 15 s, 30 s, 1 min or 5 min. The plots open at 15 s.
Timeline bar	Under the plots	Shows the whole scan. Drag the highlighted window to move through time, or click the bar to jump. Blue = following the live edge; orange = paused in the past.
Drag on a plot	Plot area	Pans back or forward in time (stops following live).
Mouse wheel on a plot	Plot area	Zooms the time window in or out around the pointer.
Hover	Plot area	Shows a readout box (top-right) with the time and the BFI/BVI values under the pointer for each side (<code>L AVG</code> , <code>R AVG</code>).
• Back to live	Top-right of the plots	Appears when you have moved into the past during a scan. Click to return to the live edge.



The window-length menu, opened from the pill at the bottom-right of the plots.

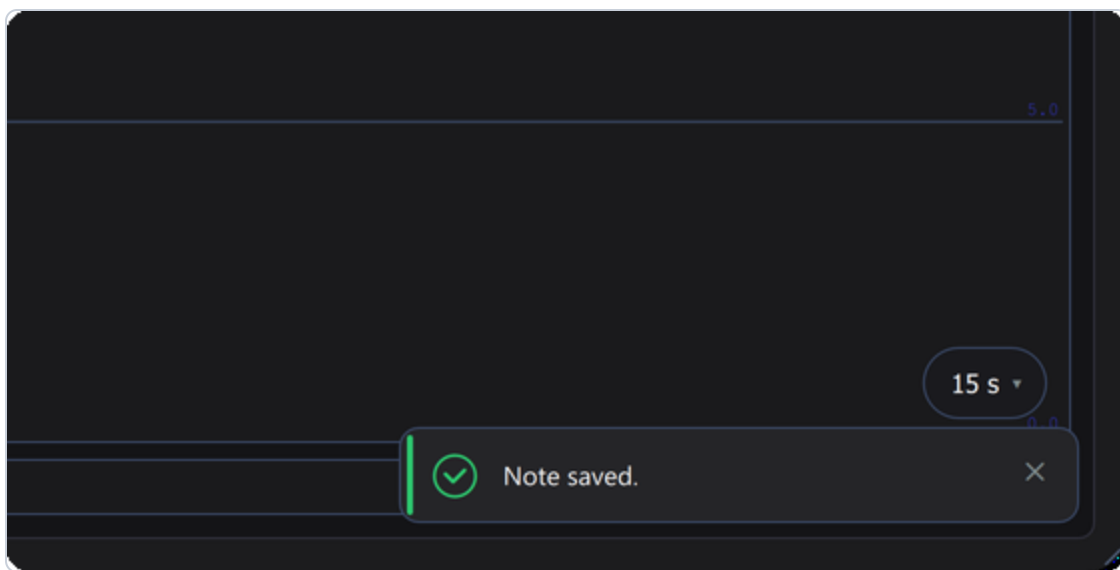
Keyboard shortcuts (active when the plot area has keyboard focus — click a plot first):

Key	Action
← / →	Pan 1 second back / forward (hold Shift to pan a full window)
+ or ↑	Zoom in
- or ↓	Zoom out
0	Reset the zoom and return to live
Home / End	Jump to the start of the scan / return to the live edge
Esc	Release keyboard focus from the plots
Space	Open Session Notes with a timestamp (during a scan)

6. Session Notes

- Open the notes with the **Notes** button at any time. They open automatically after every scan, and **Space** during a scan opens them with a timestamp line `[elapsed / clock time] -` ready for you to type after, for example `[00:04:32 / 14:32:05] -`.
- Notes are attached to a scan. During a scan they belong to that scan; after a scan they edit the scan just recorded. Each new scan starts with empty notes, so text typed before a session's first scan is cleared when that scan starts — record observations once the scan is running (Space is the quickest way) or after it ends.
- Notes are saved when the window closes — with `×`, `Esc`, or by clicking outside the window. "Note saved." confirms it. **Always close the Notes window after typing** so the text is saved.
- Saved notes appear, read-only, in Scan History (§7).

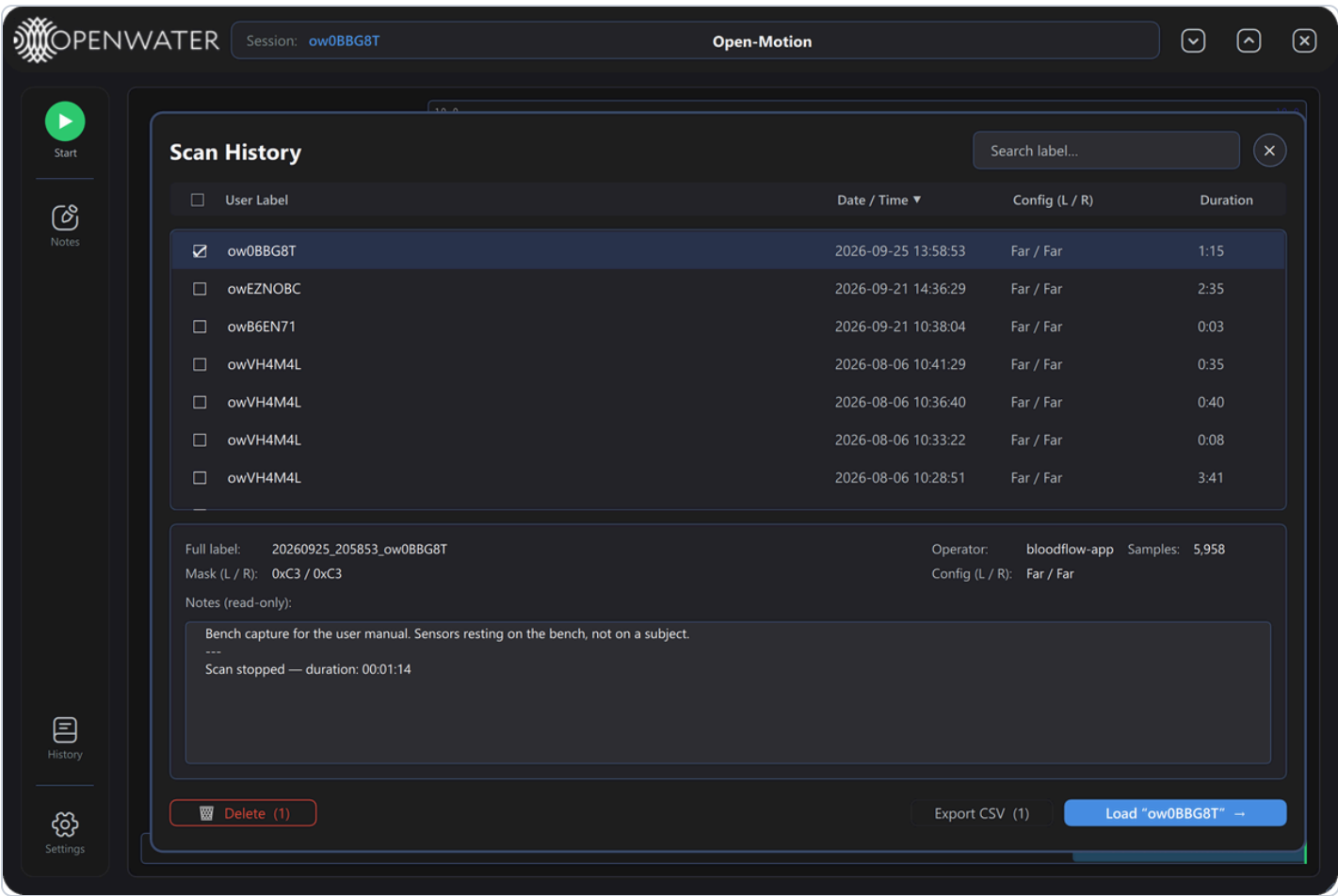
Do not enter patient-identifiable information in the notes. The same reminder is shown at the bottom of the Notes window. Notes are stored with the scan in the scan database; they are not included in exported CSV files.



Closing the Notes window saves the text.

7. Scan History

Press **History** (available when no scan is running) to review past scans.



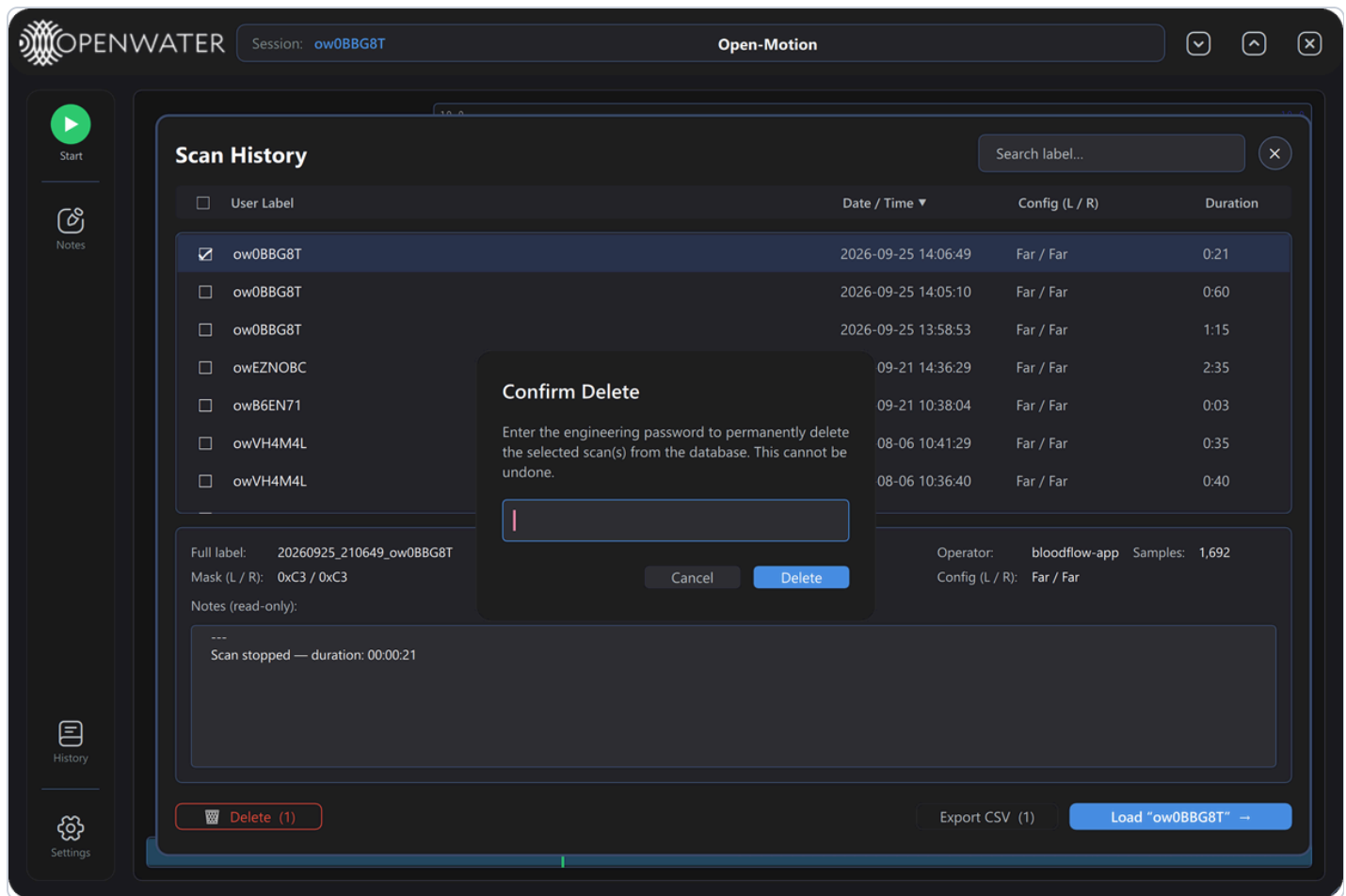
The list shows the scans recorded by the Clinical build under your Windows account, newest first.

Column / control	What it shows or does
User Label	The session label the scan was recorded under (§3). Several scans from one session share a label; use the date and time to tell them apart.
Date / Time	When the scan started, in the computer's local time.
Config (L / R)	The camera configuration used on each side (Far for clinical scans).
Duration	Scan length in minutes:seconds. – means the scan was interrupted and has no recorded end.
Search label...	Filters the list by label.
Column headers	Click to sort; click again to reverse (▲ / ▼).

Column / control	What it shows or does
Row click	Selects the scan and shows its details below: full label, operator, number of samples, camera masks, configuration and notes (read-only). The full label starts with the scan's start time in UTC, e.g. <code>20260925_205853_ow0BBG8T</code> .
Checkboxes	Mark several scans for Delete or Export. The header checkbox selects all.
Export CSV (N)	Exports the checked scans to CSV. One scan opens a Save dialog (default name <code><full label>_export.csv</code>); several scans ask for a destination folder. Interrupted scans cannot be exported.
Load "label" →	Opens the selected scan in the plot viewer for replay.
 Delete (N)	Permanently deletes the checked scans (\$7.1).
×	Closes History.

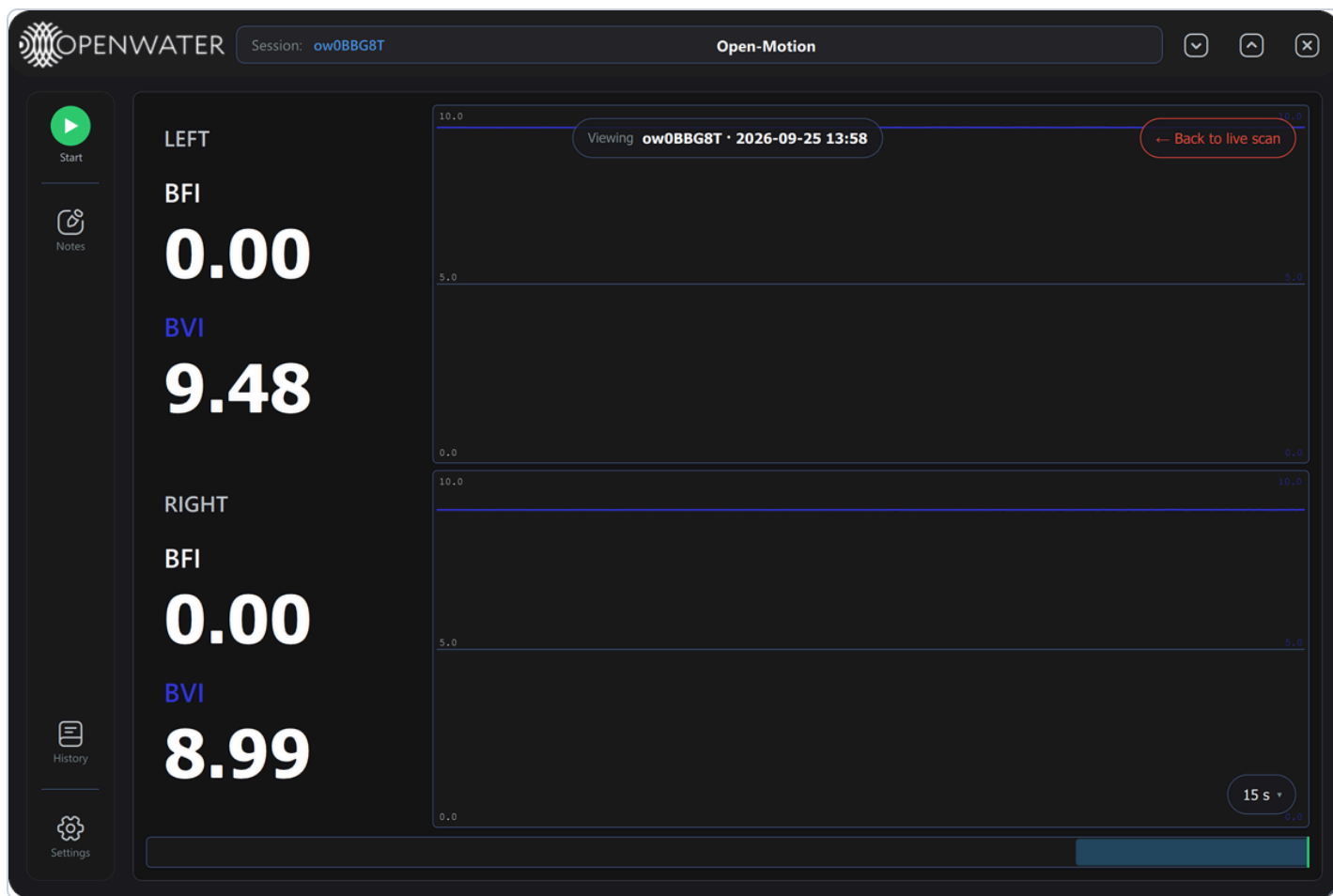
7.1 Deleting scans

Deleting is permanent and password-protected. Pressing **Delete** opens a *Confirm Delete* prompt; the scans are removed only after the password issued by Openwater is entered. Delete is unavailable while a scan is running.



7.2 Replaying a scan

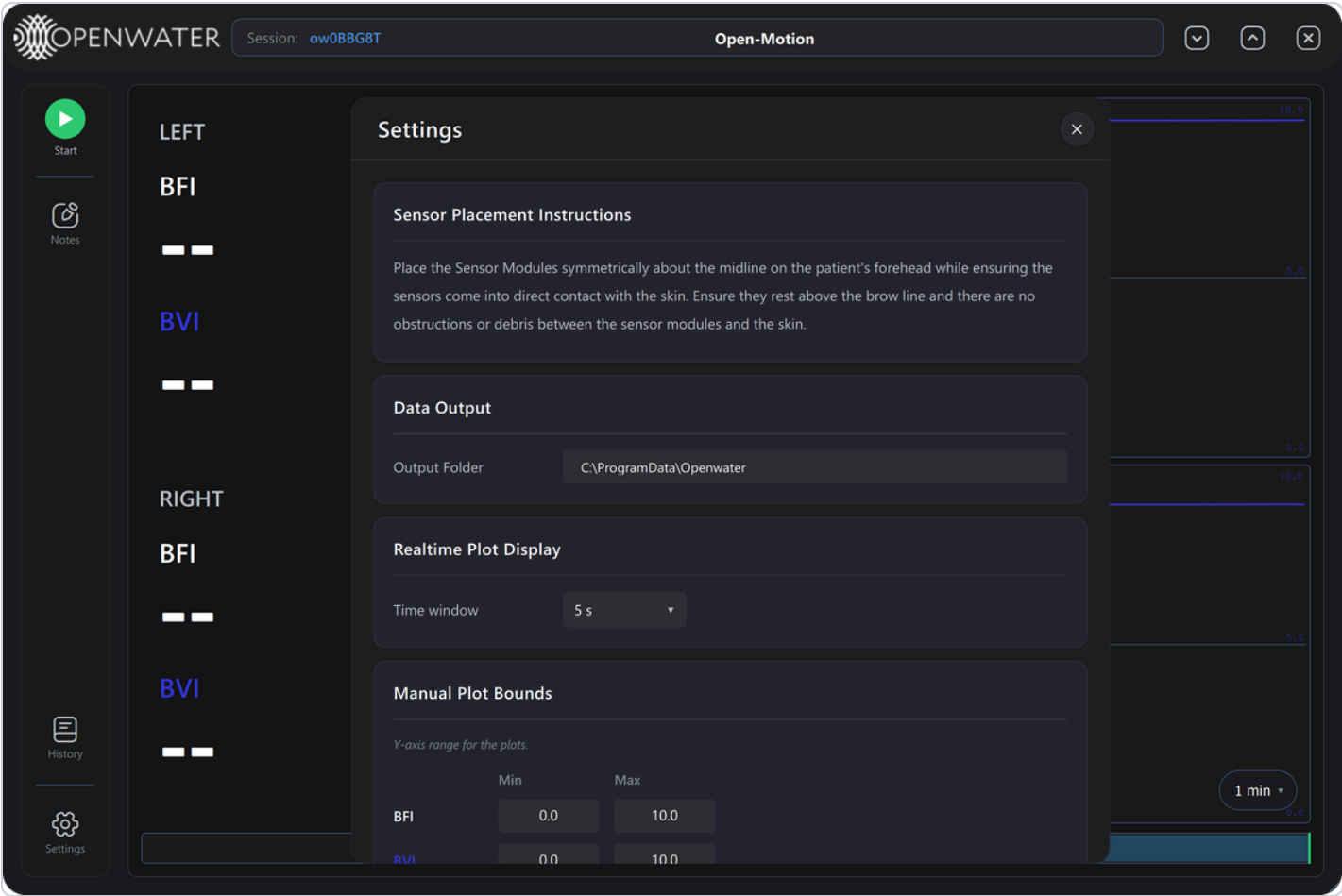
After **Load**, the plots show the recorded scan with a **Viewing** badge naming it (label · date and time). All plot navigation (§5) works the same way on a replay. When it is shown, the red ← **Back to live scan** pill returns to the live view.



A recorded scan loaded from History. The values shown come from a bench recording without a subject.

8. Settings

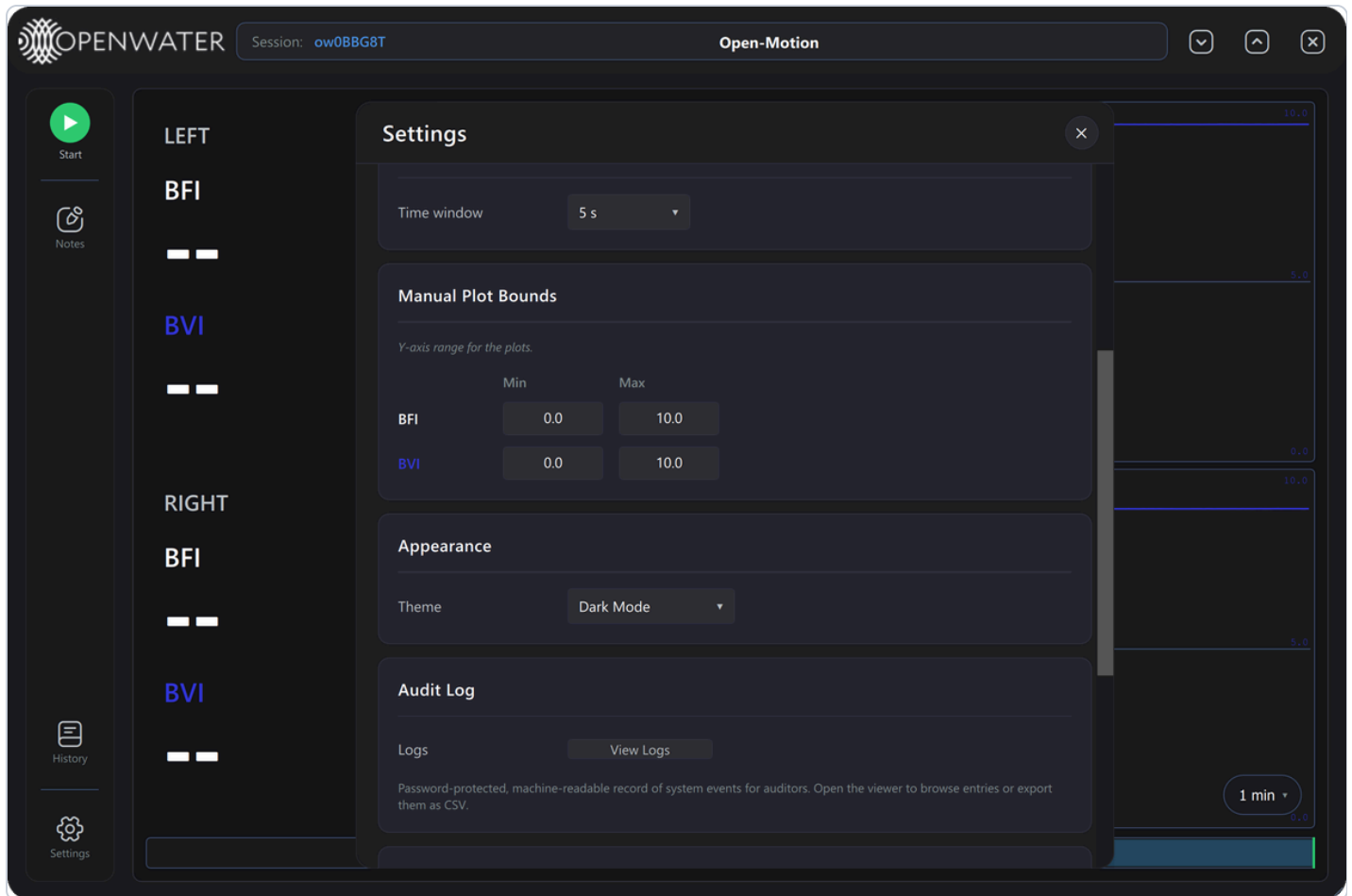
Press **Settings** (available when no scan is running). Changes apply immediately and are saved when you close Settings with **X**, **Esc**, or by clicking outside it.

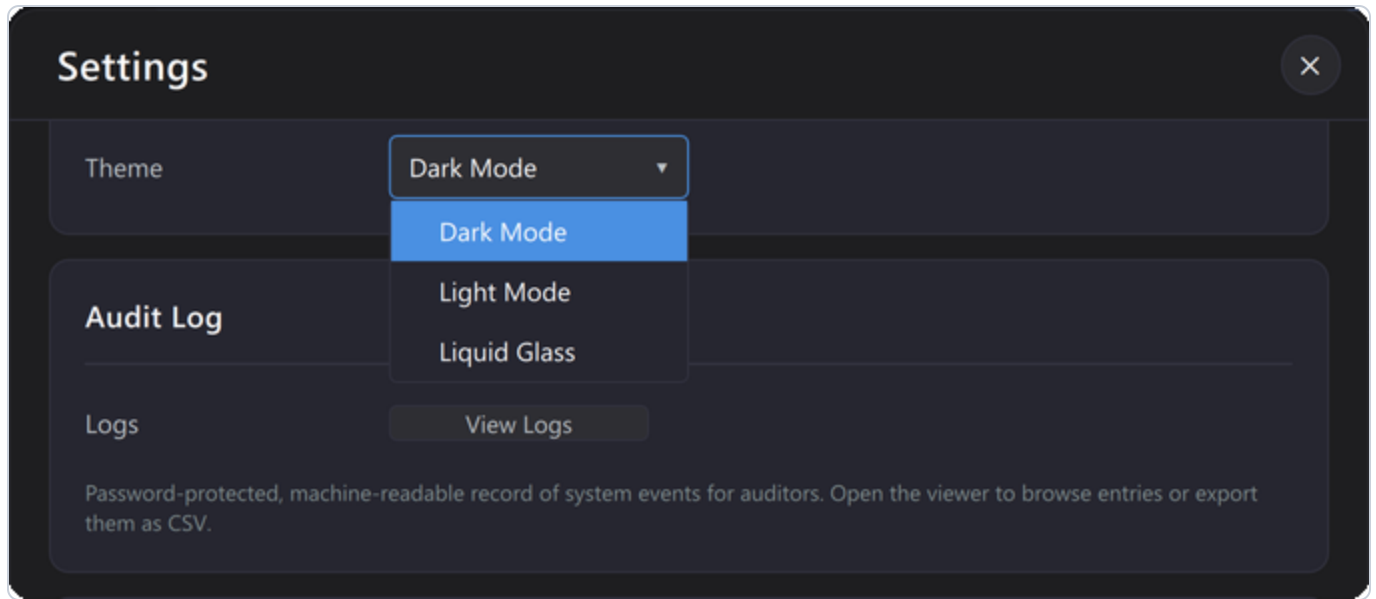


Settings, top. Captured on the 1.5.3 release candidate, which still used the shared `C:\ProgramData\Openwater` folder; a 1.5.3 installation shows your own folder under `AppData\Local\Openwater` (\$10).

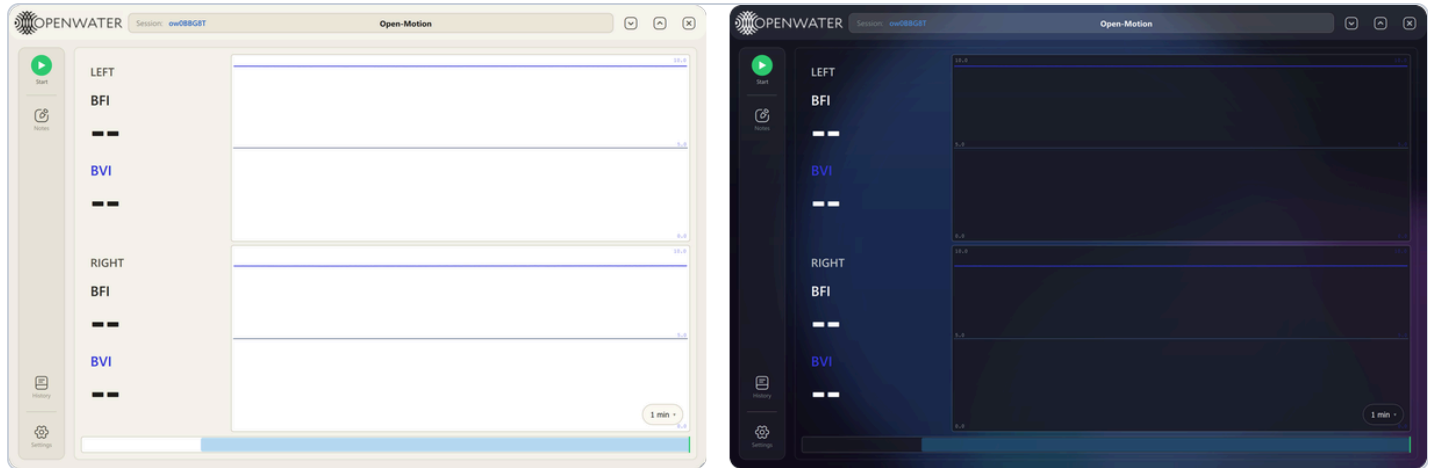
Card	Contents
Sensor Placement Instructions	The placement instructions from §4, Step 1.
Data Output	<i>Output Folder</i> : where this Windows account's scan database, logs and support files are kept. Shown for reference; it cannot be changed.
Realtime Plot Display	<i>Time window</i> : 3, 5, 15 or 30 s. To change how much time the plots show during a scan or replay, use the window-length pill under the plots (§5.2).
Manual Plot Bounds	The fixed vertical range of the plots: <i>Min</i> and <i>Max</i> for BFI and BVI, each between 0 and 10 (defaults 0.0 and 10.0). Min always stays below Max; out-of-range entries are corrected automatically.

Card	Contents
Appearance	Theme: Dark Mode (default), Light Mode , or Liquid Glass . The whole interface re-themes immediately.
Audit Log	View Logs opens the password-protected audit log (\$8.1).
About	Versions and serial numbers, plus Send Debug Logs (\$11).

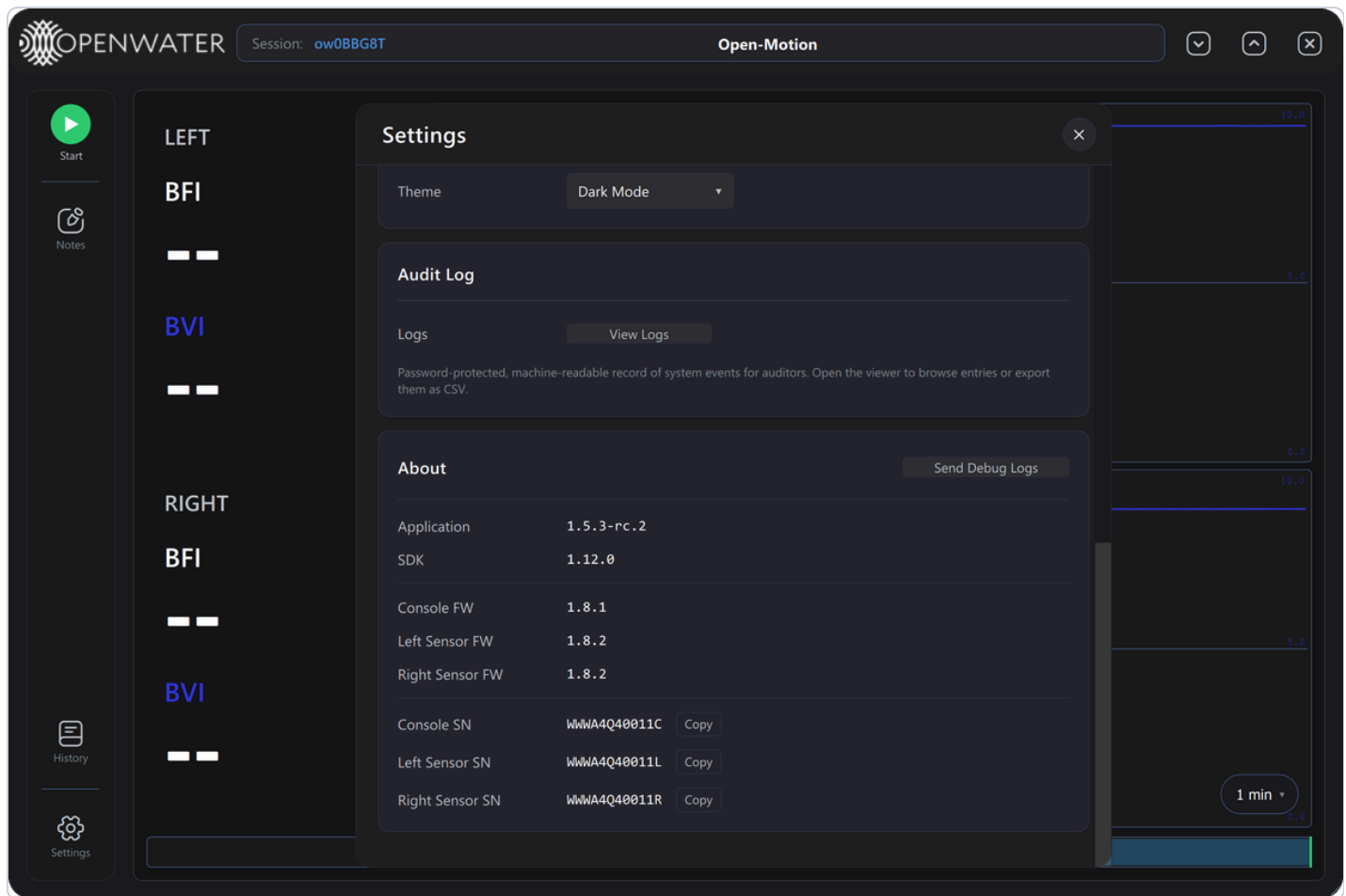




Settings → Appearance → Theme.



Light Mode (left) and Liquid Glass (right). Light Mode suits brightly lit rooms.



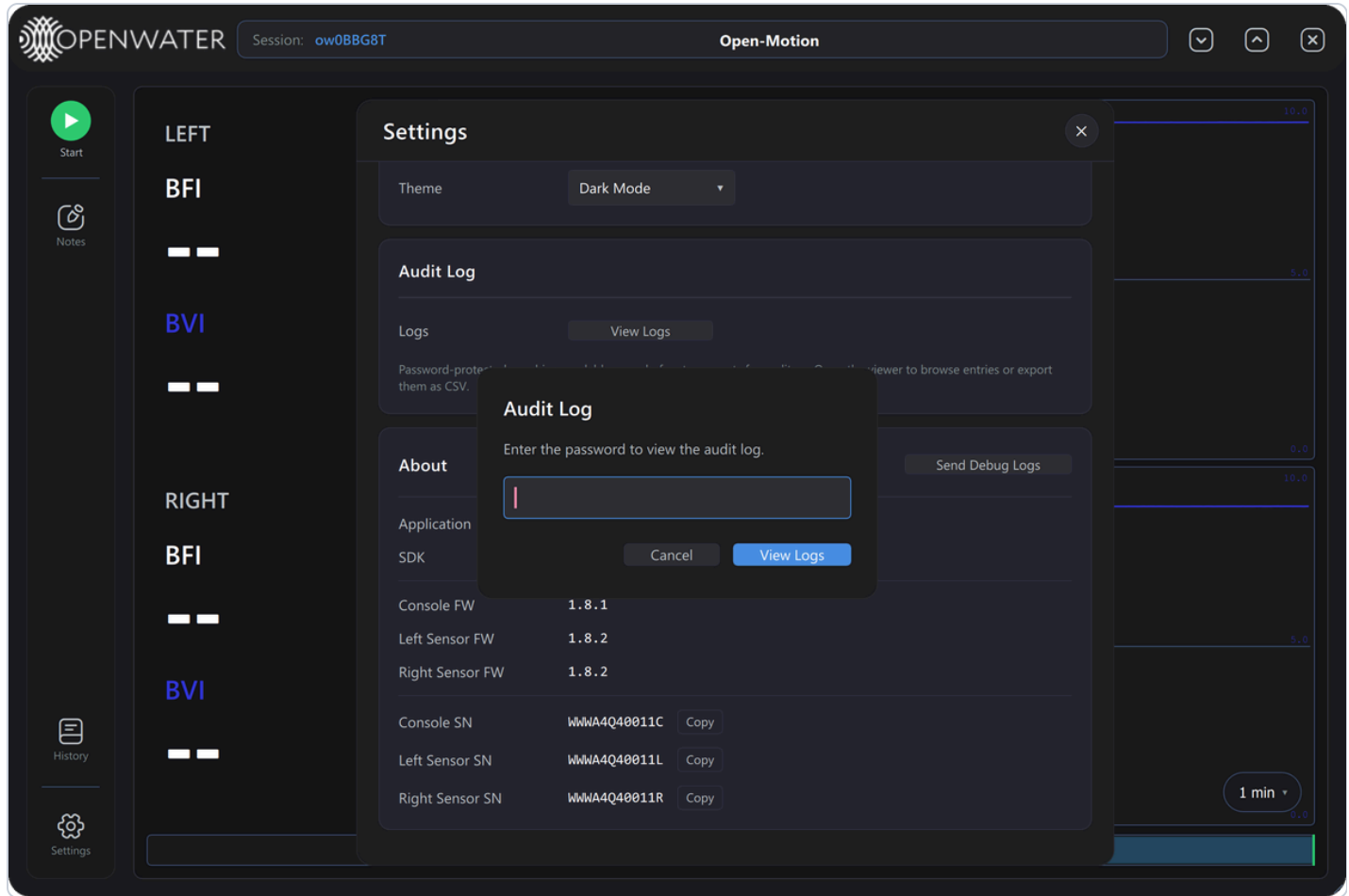
Settings → About. Captured on the 1.5.3 release candidate; a 1.5.3 installation shows `1.5.3` in the Application row.

The **About** card lists:

Row	Meaning
Application, SDK	Software versions.
Console FW, Left Sensor FW, Right Sensor FW	Firmware version of each connected device, or <i>Not connected</i> .
Console SN, Left Sensor SN, Right Sensor SN	Hardware serial numbers. Copy puts a serial number on the clipboard — include them when contacting support.

8.1 Audit Log

The audit log is a machine-readable record of system events (scans, settings changes, errors and similar) kept for auditors. **View Logs** asks for a password issued by Openwater. The viewer can filter entries by event type, date range and text, and export them as CSV.



9. Messages and errors

9.1 Messages (toasts)

Short messages appear in the bottom-right corner, color-coded: **green** = success, **yellow** = warning, **red** = error, **blue** = information. Most dismiss themselves after a few seconds (hovering over one pauses its countdown); those with an  can be closed by hand. Up to five are shown at a time.

Message	Meaning / action
"Note saved."	Session notes were stored.
"Console not detected...", "Sensor not detected...", "System not found..."	A device was not found at startup (§2.2). Check cables and power.
"Left sensor disconnected" (or <i>Right sensor, Console</i>)	A device was unplugged or powered off while no scan was running. Reconnect it.
"Could not start scan — the previous step is still finishing. Please press Start again."	The system was still busy. Wait a few seconds and press Start again.
"Scan failed: ..."	The scan could not start or run; the reason follows.
"Camera ... connection lost ...", "Camera ... reconnected ..."	A camera stopped or resumed delivering data during a scan (§4, Step 4).
"Scan ended unexpectedly ..."	See §4, Step 5.
"Laser safety warning detected. Please restart your console. If this error persists, please contact support." (red, stays on screen)	The console's hardware laser-safety monitor reported a fault. Power-cycle the console. Contact support if it recurs.
"Exported to ...", "Exported N scan(s) to ..."	A CSV export finished.
"Interrupted scans can't be exported."	The selected scan was interrupted and has no exportable data.
"Console SN copied to clipboard." (or <i>Left / Right Sensor SN</i>)	A serial number was copied from Settings → About.
"Preparing debug logs...", "Debug logs saved to ..."	See §11.

9.2 Closing while busy

See §3: pressing  during a scan or check shows "... in progress. Click X again to cancel and exit." Press  again within 5 seconds only if you intend to abandon the work.

9.3 Error dialog

Serious conditions raise a blocking **ERROR** dialog. It shows the error code (for example `E-202`), a title, an explanation, and a suggested action (→). While it is shown, the console light blinks blue.

Control	What it does
► Details	Expands technical detail, when available.
N more	Several errors are queued; they are shown one after another.
Copy details	Copies the error report to the clipboard.
Contact Support	Opens your email program with a pre-filled message to Openwater support, copies the report to the clipboard, and opens the folder containing the application log with the file selected — attach it to the email before sending.
Dismiss	Closes the dialog (or shows the next queued error). Clicking outside the dialog or pressing <code>Esc</code> does the same.

Quote the error code when contacting support.

9.4 Error code reference

Code	Title	Meaning	What to do
<code>E-101</code>	Sensor self-check failed	One or more of a sensor module's internal devices did not respond to its power-on self-check. The system cannot scan reliably.	Power-cycle the sensor module and reconnect. If it persists, the module needs service — contact support.
<code>E-102</code>	Sensor self-check unreadable	The sensor module connected but did not report its self-check result.	Power-cycle and reconnect. If it persists, contact support.
<code>E-103</code>	Console initialization failed	The console's laser-power configuration could not be applied.	Power-cycle the console and reconnect. If it persists, contact support.
<code>E-104</code>	Console not detected	Startup check: no console found (yellow message, not a dialog).	Check the console USB cable and power.
<code>E-105</code>	Camera power-on failed	The sensor module could not power on its cameras.	Power-cycle the sensor module and reconnect. If it persists, the module may need service.
<code>E-106</code>	Sensor not detected	Startup check: no sensor module found (yellow message, not a dialog).	Check the sensor USB cable and power.

Code	Title	Meaning	What to do
E-201	Laser safety monitor unresponsive	Laser safety could not be confirmed, so the laser was shut off as a precaution.	Power-cycle the system and reconnect. Do not scan until this clears. Contact support if it persists.
E-202	Laser safety trip	The laser-safety monitor tripped during a scan. The laser was shut off and the scan stops about 5 seconds later.	Remove any obstruction, let the system settle, and start a new scan. If it trips repeatedly, stop and contact support.
E-301	Scan aborted before start	A check before the laser fired failed, so the scan did not start.	Resolve the reported problem (connection, safety or configuration) and start again.
E-302	Scan could not start	The system refused a new scan, usually because the previous one is still finishing.	Wait a few seconds and try again. Reconnect if it stays busy.
E-303	Camera data lost during scan	Every camera stopped delivering data for about 3 seconds, so the scan was stopped. Data captured before the loss was saved.	Check the sensor cables and power, reconnect, and start a new scan.
E-304	Device disconnected during scan	The console or a sensor module in use was disconnected during the scan, so the scan was stopped. Data captured before the disconnection may not have been saved.	Check the USB cables and power, reconnect, and start a new scan.

10. Data storage and privacy

Each Windows account keeps its own data in a private folder:

```
C:\Users\<Windows user>\AppData\Local\Openwater
```

Location	Contents
logs\	One application log file per launch (<code>open-motion-<date>_<time>.log</code>).
data\scans.db	The scan database: all scans, their notes, and your display preferences. Encrypted in the Clinical build; the key is held by Windows for your account.
data\debug-bundles\	Zip files created by <i>Send Debug Logs</i> .

- Scan data is kept automatically. CSV files are only created when you export from Scan History (\$7). Exported files are not encrypted — store them according to your site's data-handling rules.
- Because the database key belongs to your Windows account, another Windows user cannot open your scans, and you will not see theirs.
- Scans recorded by version 1.5.2 or earlier remain in `C:\ProgramData\Openwater\data` and are not shown by this version (\$2.1).

11. Getting help

1. Open **Settings** → **About** and press **Send Debug Logs**. The application packages the last 48 hours of application logs into a zip file under `data\debug-bundles\`, opens the folder with the file selected, and shows the file's location. The package contains application logs only — no scan data.
2. Email the zip file to Openwater support at **support@openwater.health** with a description of the problem, the error code if there was one, and the serial numbers from Settings → About.
3. From an error dialog, **Contact Support** prepares the email for you (§9.3).

12. Known issues in 1.5.3

Issue	Workaround
The "Debug logs saved..." message names <code>support@openwater.cc</code> .	Send debug logs to the support address in §11.
Plot data loads slowly when zooming out to the beginning of a long scan.	Allow a moment for the plot to fill in.
A scan just under a whole minute can show a duration such as <code>0:60</code> in Scan History.	Read it as <code>1:00</code> . The duration line in the scan's notes is correct.
When the window is maximized, the restore button shows an incorrect icon.	The button still restores the window.